We introduced the concept of the stateless server in part 3 of this book and explained how to use decoupling in chapter 13. In this chapter, you'll return to the concept of the stateless server and also work through an example of synchronous and asynchronous decoupling.

17.1 Managing a dynamic EC2 instance pool

Imagine that you need to provide a scalable infrastructure to run a web application, such as a blogging platform. You need to launch uniform virtual machines when the number of requests grows and terminate virtual machines when the number of requests shrinks. To adapt to the current workload in an automated way, you need to be able to launch and terminate VMs automatically. Therefore, the configuration and deployment of the web application needs to be done during bootstrapping, without human interaction.

In this section, you will create an Auto Scaling group. Next, you will learn how to change the number of EC2 instances launched by the Auto Scaling group based on scheduled actions. Afterward, you will learn how to scale based on a utilization metric provided by CloudWatch. Auto Scaling groups allows you to manage such a dynamic EC2 instance pool in the following ways:

- Dynamically adjust the number of virtual machines that are running
- Launch, configure, and deploy uniform virtual machines

The Auto Scaling group grows and shrinks within the bounds you define. Defining a minimum of two virtual machines allows you to make sure at least two virtual machines are running in different availability zones to plan for failure. Conversely, defining a maximum number of virtual machines ensures you are not spending more money than you intended for your infrastructure. As figure 17.2 shows, autoscaling consists of three parts:

- A launch template that defines the size, image, and configuration of virtual machines
- An Auto Scaling group that specifies how many virtual machines need to be running based on the launch template
- Scaling plans that adjust the desired number of EC2 instances in the Auto Scaling group based on a plan or dynamically